FastAPI Question & Answers

1.What is FastAPI?

CRUD stands for Create, Read, Update, and Delete. These are the four basic operations used in database applications. In FastAPI, CRUD helps manage data like students, employees, or products. Each operation is handled using different API methods. It makes data management simple and organized.

2. Why do we use the POST method for inserting data?

The POST method is used to send new data to the server. It helps create a new record in the database. The data is usually sent through the request body in JSON format. FastAPI reads and validates this data easily. POST is mainly used for adding new entries.

3. Difference between PUT and PATCH

PUT is used to completely update an existing record. It replaces all the fields with new data. PATCH is used to update only selected fields of a record. PUT needs full data, while PATCH needs only changed values. PATCH is more flexible for small updates.

4. What is the path parameter?

A path parameter is a value passed inside the URL path. It is used to identify a specific resource like a student ID. For example, in /students/101, 101 is the path parameter. FastAPI automatically reads this value from the URL. It helps access specific records easily.

5. What is the request body in FastAPI?

The request body contains the data sent by the client to the API. It is usually in JSON format for creating or updating records. FastAPI uses it mainly with POST, PUT, and PATCH methods. The data is validated using Pydantic models. It helps send structured information to the server.

6. Why do we use Pydantic models?

Pydantic models are used for data validation in FastAPI. They define the structure and data types of input and output data. This helps ensure only correct data is accepted by the API. It reduces errors and improves code quality. It also makes the code cleaner and easier to understand.

7. What status code is returned for successful deletion?

The status code for successful deletion is usually 200 OK or 204 No Content. Code 200 means the deletion was successful with a response message. Code 204 means deletion was successful without returning data. Both confirm the record was removed properly. It depends on how the API is designed.

8. How can we check whether a student exists before updating?

We first search for the student ID in the database or list. If the student is found, the update process continues. If the student does not exist, an error message is returned. FastAPI can raise an HTTP exception for this. This avoids updating wrong or missing records.

9. What happens if duplicate IDs are inserted?

Duplicate IDs can cause confusion and data inconsistency in the system. It becomes difficult to identify the correct record. Usually, the API should check existing IDs before inserting. If a duplicate is found, it should return an error message. This helps maintain unique and correct data.

10. How does FastAPI automatically generate Swagger UI?

FastAPI automatically generates Swagger UI from the API code. It reads route definitions, request types, and Pydantic models. This creates interactive API documentation without extra coding. Users can test APIs directly from the browser. It is available by opening the /docs URL.